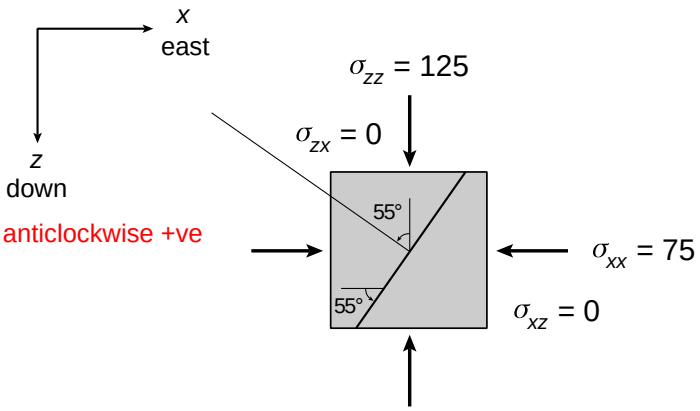
Exercise 4 solution

If the state of stress at a point is σ_{xx} = 75 MPa, σ_{zz} = 125 MPa, and σ_{zx} = 0, where z is depth, and the x axis points eastwards, what are the normal and shear stresses on a fault plane oriented 000/55W that passes through the same point? What sense of slip might be expected? Given that the orientation of the x and z axes have been defined as they are, why is the strike of the fault important in this question?

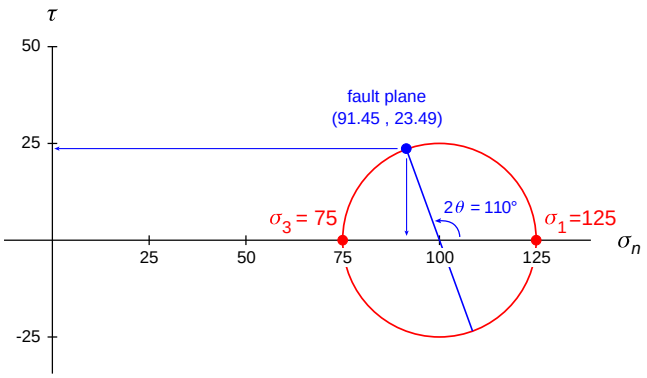


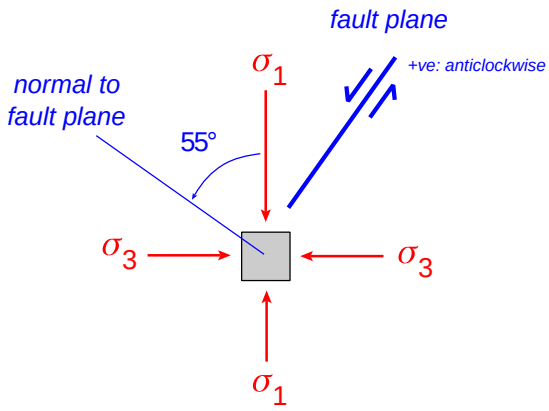
$$\begin{bmatrix} 75 & 0 \\ 0 & 125 \end{bmatrix}$$

σ_{zx} is zero and so σ_{zz} and σ_{xx} are principal stresses

$\sigma_{zz} > \sigma_{xx}$ and so $\sigma_{zz} = \sigma_1$

$\sigma_{xx} = \sigma_3$





Or if you prefer to use the formulae:

$$\sigma_n = \frac{(\sigma_1+\sigma_3)}{2} + \frac{(\sigma_1-\sigma_3)}{2}\cos 2\theta \quad ; \quad \tau = \frac{(\sigma_1-\sigma_3)}{2}\sin 2\theta$$

With

$$\sigma_1 = 125 \text{ MPa} \quad ; \quad \sigma_3 = 75 \text{ MPa} \quad ; \quad \theta = 55°$$

then

$$\sigma_n = (200/2) + [(50/2)\cos(110)] = 91.45 \text{ MPa}$$

$$\tau = (50/2)\sin(110) = 23.49 \text{ MPa}$$

σ_1 is vertical and so from the Andersonian picture of faulting we expect the sense of slip on the fault to be normal. Note that this sense of slip also follows from the sign of the shear stress on the fault plane. The strike of the fault is important because it means that the fault plane contains the intermediate principal stress, σ_2, and so we can treat this three dimensional stress problem as a two dimensional one.